21 The time period, T seconds, of a simple pendulum of length l cm is given by the formula

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Katie uses a simple pendulum in an experiment to find an estimate for the value of g.

Here are her results.

$l = 52.0$ correct to 3 significant figures.
$T = 1.45$ correct to 3 significant figures.

Work out the upper bound and the lower bound for the value of g.
Use $\pi = 3.142$
You must show all your working.

$T = 2\pi\ sqrt(l/g) \Rightarrow g = 4\pi^2 l / T^2.$

$51.95 \le l < 52.05;\ 1.445 \le T < 1.455.$

Upper g: largest l, smallest T.

$g < 4(3.142)^2(52.05)/(1.445)^2 = 984.367...$

Lower g: smallest l, largest T.

$g > 4(3.142)^2(51.95)/(1.455)^2 = 969.018...$

1 mark
correct input bounds

1 mark
opposite extremes used

1 mark
$g=4\pi^2 l/T^2$

1 mark
both final bounds

upper bound = 984.3678

lower bound = 969.0182

(Total for Question 21 is 4 marks)

TOTAL FOR PAPER IS 80 MARKS

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